

1. Purpose

- 1.1. The purpose of this test is to evaluate the mechanical durability of the device when subjected to repeated flexing. The test simulates the mechanical stress experienced by the device during normal handling and use.
- 1.2. This evaluation helps determine whether the device maintains structural integrity, functional performance, and safe performance after repeated mechanical deformation.

2. Scope

- 2.1. This protocol applies to fatigue testing of the device development prototypes and ITA prototypes to be used in the initial clinical study.

3. Description

- 3.1. This test evaluates the mechanical durability of the device when subjected to repeated bending. Using the test fixture, the device is flexed upward and downward to a 35° angle for 3,250 cycles, simulating its expected lifetime stress. Devices are positioned in a consistent orientation to ensure repeatable loading, and functional performance (heating and TENS output) is verified before and after the bending cycles.

3.2. Material and Equipment

Test Specimen	Description
Device	Tested device
Equipment & Instruments	
Thermocouple probe	Calibrated
Thermometer	Compatible with thermocouple probes. Calibrated.
Timer	Calibrated
Oscilloscope	Calibrated
Tools, materials and fixtures	
Test Fixture	Test fixture used to bend the device multiple times.

3.3. Sample Size

- 3.3.1. The sample size for this test is 10 devices, which will be tested to provide an initial assessment of the product's mechanical robustness under repeated bending conditions. This test aims to identify potential failure modes that may occur during normal use and to evaluate the device's durability when subjected to cyclic flexion over its expected service life.

3.4. Procedure

3.4.1. Pre-Test Functional Verification:

- 3.4.1.1. Position two calibrated thermocouples in the flat front face of the device.
- 3.4.1.2. Power on the device and run it at the maximum level of heat. Wait until the heating element reaches its target temperature on both thermocouples ($42.5\text{ }^{\circ}\text{C} \pm 0.5\text{ }^{\circ}\text{C}$). Use the timer to record the time required to reach this temperature.
- 3.4.1.3. Once both of the thermocouple readings reach $42.5\text{ }^{\circ}\text{C}$ continue monitoring for at least 2 minutes to ensure the temperature remains stable and does not overshoot the target range.
- 3.4.1.4. Record the time required to reach $42.5\text{ }^{\circ}\text{C}$ and the maximum temperature observed.
- 3.4.1.5. Follow the procedure of the TENS performance test in Appendix C to verify that the TENS is working correctly.
- 3.4.1.6. Repeat steps 3.4.1.1 through 3.4.1.5 for all samples.

3.4.2. Fatigue Test

- 3.4.2.1. Place the device facing upward in the test fixture as shown in image 1.



Image1. Placement of device in fixture

- 3.4.2.2. Clamp the sides of the device and secure them to the test fixture as shown in image 2.



Image 2. Clamping the device

- 3.4.2.3. Press the start button on the fixture controller to begin the motorcycle. The fixture will bend the device upward and downward to the programmed angle of 35° . The programmed test sequence will automatically stop after completing 3250 bending cycles. The rationale for selecting the bending angle and the total number of cycles is provided in Annex A and Annex B, respectively.

- 3.4.2.4. Repeat steps 4.4.2.1 through 4.4.2.3 for all samples.

3.4.3. Post-Test Inspection:

- 3.4.3.1. Perform a visual inspection of the device. Search that neither the heat element or the TENS have been detached and that there are no major deformations on the device. If any defects are observed, describe them in WS.TP.010.20.
- 3.4.3.2. Follow steps 4.4.1.1 through 4.4.1.5 to perform a post-test inspection of the heat element and TENS.

4. Acceptance Criteria

- 4.1. No detachment of the heating element or TENS from the assembly.
- 4.2. No cracks, fractures, or visible damage to structural components.
- 4.3. Post-test time to reach 42.5°C shall not exceed the pre-test value by more than 5%.
- 4.4. Post-test maximum temperature shall remain within the allowable tolerance (42.5 °C ± 0.5 °C).
- 4.5. Post-test TENS output amplitude at selected power level shall remain within ±20% of the pre-test measurement.
- 4.6. Post-test TENS output current at selected power level shall remain within ±20% of the pre-test measurement.
- 4.7. Post-test TENS output frequency shall remain within ±20% of the pre-test measurement.

5. References

- 5.1. NA

6. Appendices

Appendix A. Bending Angle Justification

To estimate a realistic bending angle, the frontal section of the abdomen was modeled as part of a circle. The values used were population averages for an adult woman abdomen dimension:

- Average abdominal depth (front-to-back): 215 mm.
- Abdominal width used: $W = 360$ mm.

Because the device covers only the frontal portion of the abdomen, the device sagitta is approximated as one quarter of the abdominal depth:

$$s = 215/4$$

$$s = 53.75 \text{ mm}$$

From circle geometry, the radius R corresponding to a sagitta s and chord width “ W ” is:

$$R = \frac{s^2 + (\frac{W}{2})^2}{2s}$$

$$R = \frac{53.75^2 + (\frac{360}{2})^2}{2 \cdot 53.75}$$

$$R \approx 328$$

This means that if a circle is drawn matching the curvature that the device conforms to on an average woman, its radius would be approximately 328 mm. The bending angle θ at the device's sides can then be estimated using:

$$\theta = 2\arcsin\left(\frac{W_D}{2R}\right) * \frac{180}{\pi}$$

Where:

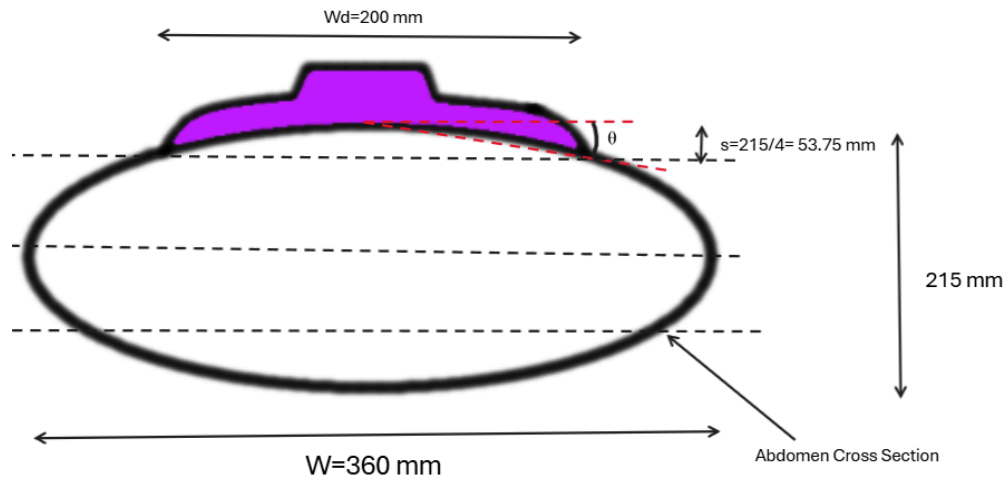
$$W_D = \text{device width}$$

$$R = \text{radius}$$

$$\theta = 2\arcsin\left(\frac{200}{2*328}\right) * \frac{180}{\pi}$$

$$\theta \approx 35^\circ$$

Thus, the estimated bending angle for the device in this representative scenario is approximately 35° . In image 3 you can observe a visual representation of the model made for obtaining the bending angle.



$$R = \frac{s^2 + \left(\frac{W}{2}\right)^2}{2s} \quad \theta = 2\arcsin\left(\frac{W_D}{2R}\right) * \frac{180}{\pi}$$

$$R = \frac{53.75^2 + \left(\frac{360}{2}\right)^2}{2*53.75} \quad \theta = 2\arcsin\left(\frac{200}{2*328}\right) * \frac{180}{\pi}$$

$$R \approx 328 \quad \theta \approx 35^\circ$$

Image 3. Modelling of the device to obtain the bending angle.

Annex B. Justification of Total Number of Bends

The following calculations and rationale were used to determine the total number of bending cycles to be applied on the device. The following are the planned used conditions for the device:

- Uses per day during symptomatic period: 5
- Average symptomatic days per menstrual cycle: 5
- Average menstrual cycle length: 28 days (≈ 13 cycles/year)
- Service life limited by battery: 2-5 years (5 years used)
- Safety factor: $\times 2$

For the purpose of this calculation, one “use” is equivalent to one complete bending cycle (one upward and one downward deflection to the specified $\pm 35^\circ$). This reflects the flexion imposed on the device during handling. The following formula can be used to estimate the number of times the device is bent during its expected lifetime.

$$\text{Total bends} = \text{Uses per day} * \text{days per cycle} * \text{cycles per year} * \text{expected lifetime} * \text{safety factor}$$

$$\text{Total bends} = 5 * 5 * 13 * 5 * 2$$

$$\text{Total bends} = 3250 \text{ cycles}$$

The number of bend cycles that will be used for the protocol is 3250 cycles.

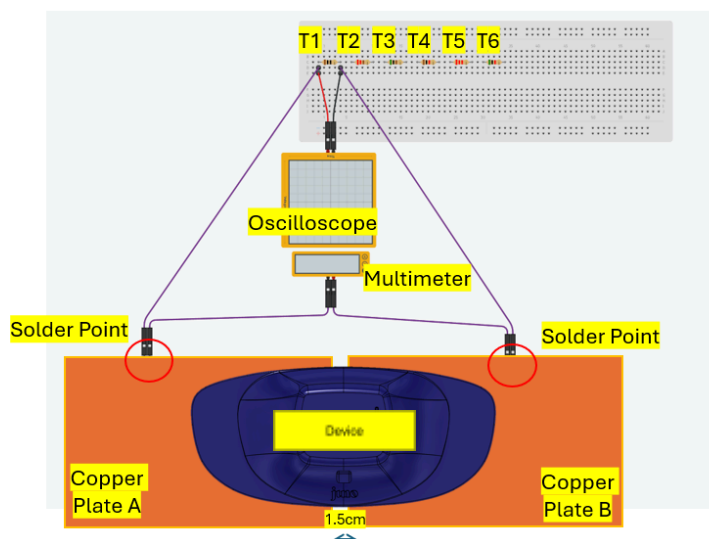
Appendix C. TENS performance Test

1. Procedure

1.1. General Conditions

- 1.1.1. Place the device electrodes on the copper plates (6×6 in.).
- 1.1.2. Connect copper plates to the test load ($500\Omega \pm 1\%$ 1W) via protoboard and jumpers.
- 1.1.3. Connect oscilloscope probes and multimeter across the load.
- 1.1.4. Ensure all instruments are calibrated and zeroed before measurements.

TENS PERFORMANCE SET UP



1.2. Baseline Measurements

- 1.2.1. Impedance Check (without device):
 - 1.2.1.1. Connect the test load ($500\Omega \pm 1\%$ 1W).
 - 1.2.1.2. Measure and record baseline impedance with oscilloscope/multimeter.
- 1.2.2. Impedance Check (with device, TENS OFF):
 - 1.2.2.1. Place the device on the copper plates, power it ON, but keep TENS system off.
 - 1.2.2.2. Measure and record impedance again for the test load using a calibrated multimeter in resistance mode.

1.3. TENS MODE MEASUREMENTS

- 1.3.1. Set the device to Tens Mode 2.
- 1.3.2. Set the device to Tens Level 5.
- 1.3.3. Measure the voltage amplitude. Voltage amplitude shall be measured as peak-to-peak (V_{pp}) over at least 10 consecutive pulses using oscilloscope cursors or automated measurement functions. Record Theoretical current ($I=V/R$) during the pulse.
- 1.3.4. Record Frequency HZ from oscilloscope Trace.
- 1.3.5. Record an Oscilloscope image.
- 1.3.6. Complete the Table:

Theoretical value Ω	Measured Value Ω	Voltage (V)	Current (mA)	Frequency (Hz)	Image